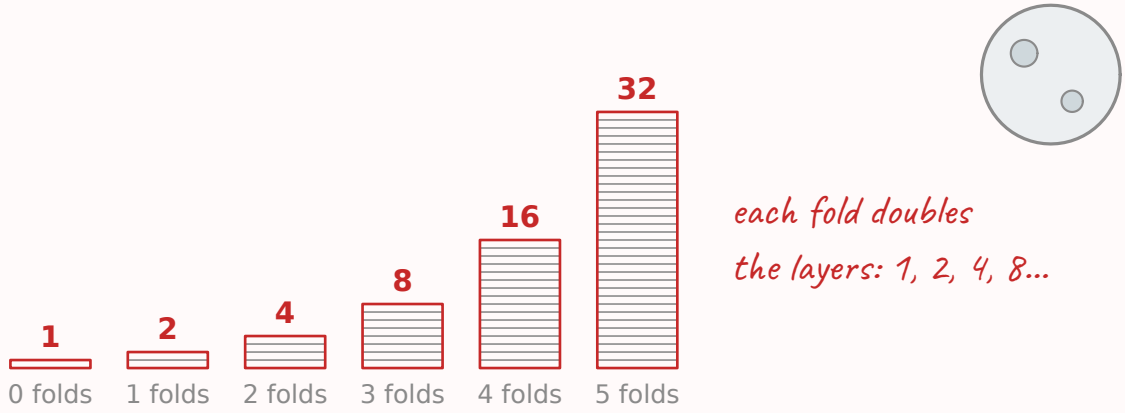
Chapter 21

The exponential function

fold a sheet of paper 42 times and reach the Moon



Most growth you know is **adding**: a taxi fare goes up £2 per mile. **Exponential** growth is **multiplying**: the amount doubles (or grows by 5%, or halves) at every step. It starts slowly and then explodes, which is why it surprises everyone.

You'll see it race against x^2, meet its mirror image (decay), and watch the famous number $e = 2.718\ldots$ appear out of a bank account. Then two real decays from your lectures: cooling and the current in a circuit.

Goes with **Lecture 7** (power vs exponential, growth vs decay, the number e).

Folding paper: multiply, multiply, multiply

A sheet of paper is about 0.1 mm thick. Each fold **doubles** the thickness. After n folds there are 2^n layers:

folds n	layers 2^n	thickness	about as tall as
0	1	0.1 mm	a sheet
10	1 024	102 mm	a paperback book
20	1 048 576	105 m	a 30-storey tower
30	1 073 741 824	107 km	into space
42	4 398 046 511 104	440,000 km	past the Moon (384,000 km)

(Nobody can actually fold paper 42 times; about 7 is the limit. But the **maths** is the point.) The thickness is 0.1×2^n mm. Notice where the letter is: **up in the power**. That's what makes a function exponential.

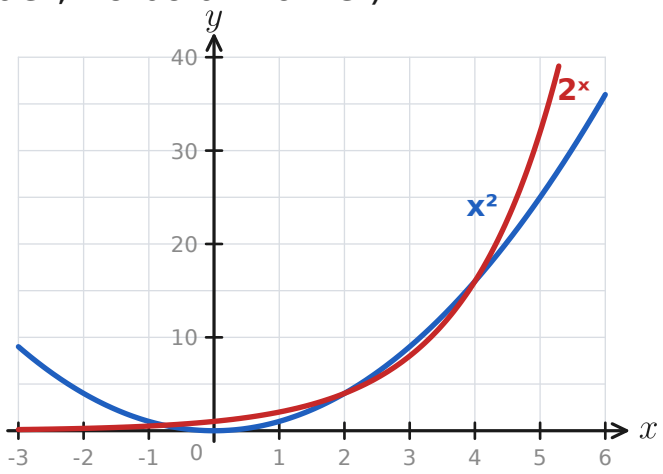
Chapter 17 link: the layers 1, 2, 4, 8, ... are a geometric progression with $r = 2$. An exponential function is a G.P. that's allowed to take in-between inputs too.

The race: x² against 2ˣ

They look similar, but in x^2 the letter is the **base** (a power function, Chapter 19) and in 2^x the letter is the **power** (exponential). Lecture 7 races them:

x	0	1	2	3	4	5	6	8	10	20
x²	0	1	4	**9**	16	25	36	64	100	400
2ˣ	**1**	**2**	4	8	16	**32**	**64**	**256**	**1024**	**1.05M**

They **tie twice**: at $x = 2$ (both are 4) and at $x = 4$ (both are 16). Between those, x^2 is ahead ($3^2 = 9$ beats $2^3 = 8$). After $x = 4$, 2^x runs away: at $x = 20$ it's over a million while x^2 is only 400. (Bold = the leader; no bold = a tie.)



Every step right, x^2 adds a bit more (+5, then +7, then +9…).

2^x DOUBLES every step. Doubling always wins in the end.

On the left, 2^x never goes negative or reaches 0: 2 to the −3 = 1/8.

The "where's the x?" test: letter in the base → power function. Letter in the power → exponential.

The shape: growth and its mirror, decay

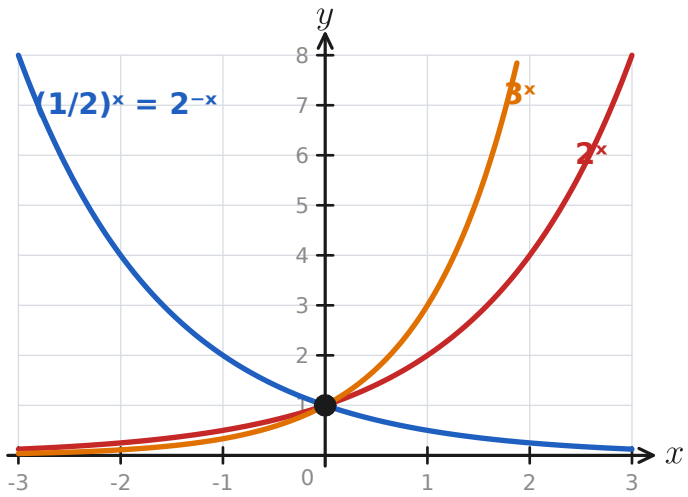
RULE: exponential graphs

$y = A \times b^x$ ($b > 0$): the graph passes through $(0, A)$, because $b^0 = 1$. A is the **starting value**.

$b > 1$: **growth**. $0 < b < 1$: **decay**. A bigger b grows faster.

$y = Ae^{kx}$: $k > 0$ grows, $k < 0$ decays; again A is the value at $x = 0$.

The x-axis is an **asymptote**: $A \times b^x$ (with $A > 0$) is always positive, never 0.



Every one passes through (0, 1), because anything to the power 0 is 1 (Chapter 9).

Base bigger than 1: GROWTH.

Bigger base: steeper.

Base between 0 and 1: DECAY.

$(1/2)^x$ is 2^x reflected in the y-axis.

All of them hug the x-axis on one side without ever touching it: the x-axis is an **asymptote** (Chapter 19). Lecture 7: growth models savings (and student debt!); decay models radioactivity and electric currents dying away.

Start amount × multiplier to the power of steps

RULE: percentage growth and decay

An increase of $r\%$ per step multiplies by $1 + \frac{r}{100}$. A decrease of $r\%$ multiplies by $1 - \frac{r}{100}$ (Chapter 4's multipliers).

After n steps: **amount = start × multiplier^n**, i.e. $y = A \times b^n$.

Compound interest, rate r (as a decimal) per year, added n times a year, for t years: $A\left(1 + \frac{r}{n}\right)^{nt}$. Each payment adds $\frac{r}{n}$, and there are nt payments.

Real exponentials are written $y = A \times b^x$: A is the **starting amount**, b is the **multiplier per step** (Chapter 4's multipliers), and x counts the steps.

$$y = A \times b^x$$

Savings: £1000 at 5% interest a year. Each year the money is multiplied by 1.05:

$$1000 \times 1.05 \times 1.05 \times \cdots = 1000 \times 1.05^t$$ ← *t years of × 1.05*

$$1000 \times 1.05^{10} \approx 1628.89$$ ← *after 10 years: £1628.89*

Compare simple interest (£50 added every year, a straight line): £1500 after 10 years. The extra £128.89 is interest earning interest.

Decrease: a £15,000 car loses 15% of its value a year. Multiplier $1 - 0.15 = 0.85$:

$$15000 \times 0.85^3 \approx 9211.88$$ ← *value after 3 years: £9211.88*

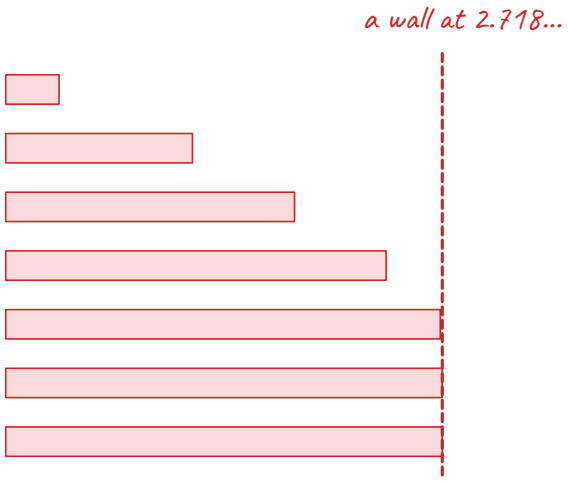
Interest added monthly: £1000 at 6% a year, paid monthly for 2 years: $n = 12$, so each month adds $\frac{0.06}{12} = 0.005$, and there are $12 \times 2 = 24$ payments:

$$1000\left(1 + \frac{0.06}{12}\right)^{24} = 1000 \times 1.005^{24} \approx 1127.16$$

The number e falls out of a bank account

A generous bank offers **100% interest** per year on £1. Paid once at the end: £2. But what if they pay **half** of it (50%) every six months, so the first half starts earning too? Or a quarter every three months? Paying n times a year multiplies by $(1 + \frac{1}{n})$, n times:

paid n times	$(1 + 1/n)^n$
1 (once a year)	£2.00000
2 (every 6 months)	£2.25000
4 (every 3 months)	£2.44141
12 (monthly)	£2.61304
365 (daily)	£2.71457
8 760 (hourly)	£2.71813
1 000 000 (a million times)	£2.71828



More often helps less and less. It never passes a ceiling of about **2.71828**. That number is called e (Lecture 7: $e = (1 + \frac{1}{n})^n$ as $n \to \infty$). Like π, nobody invented it and its decimals never end or repeat.

The function e^x is **the** exponential function; you'll also see it written $\exp(x)$. Your calculator has an e^x button (usually shift + ln). It behaves just like 2^x or 3^x, sitting between them, and it's the one nature and calculus prefer (Chapter 34 shows why).

Real decay: cooling tea and a draining capacitor

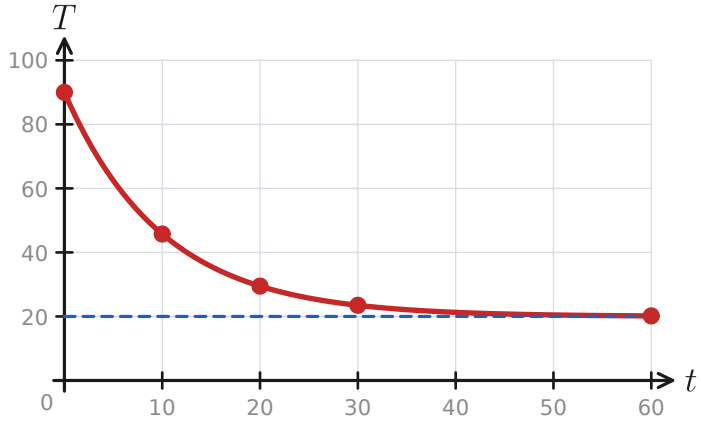
RULE: decay towards a level

$y = A + Be^{-kt}$ (with $k > 0$) starts at $y = A + B$ when $t = 0$ and **levels off at $y = A$**: as t grows, $e^{-kt} \to 0$. The line $y = A$ is a horizontal asymptote.

In $y = Ie^{-t/\tau}$, the **time constant** τ is the time to fall to $e^{-1} \approx 37\%$ of the start. Every further τ multiplies by 0.37 again.

A mug of tea at 90 °C cools in a 20 °C room. The **difference** from room temperature (70 °C to start) decays exponentially:

$$T = 20 + 70e^{-0.1t}$$ ← *t in minutes*



t = 0: 20 + 70 × 1 = 90 °C
t = 10: 20 + 70/e ≈ 45.8 °C
t = 20: ≈ 29.5 °C
t = 30: ≈ 23.5 °C

It creeps towards 20 °C (blue asymptote) but never gets there.

The capacitor (Lecture 7, Question 2). A current starts at I and dies away:

$$i = Ie^{-t/CR}$$ ← *C and R are the circuit's capacitance and resistance*

When $t = CR$ the power is -1, so $i = Ie^{-1} \approx 0.37I$: the current has dropped to 37%. CR is called the **time constant**. After another CR it's 37% of that, and so on: the same fraction lost in every equal stretch of time. That's the fingerprint of every exponential decay.

Chapter 22 answers the reverse question, "how **long** until the current is 10%?", which needs logarithms.

Find the unknown: the model from the data

PROCEDURE: finding A and b from two points

1. Each point (x, y) on $y = A \times b^x$ gives an equation: put in x and y.
2. If one point has $x = 0$, it gives A straight away ($b^0 = 1$).
3. Otherwise **divide** one equation by the other: A cancels and leaves a power of b. Take the root (use the positive one: $b > 0$).
4. Put b back to find A, and check with the other point.

Example: $y = A \times b^x$ passes through $(0, 3)$ and $(2, 12)$. Find A and b.

$$3 = A \times b^0 = A \longleftarrow x = 0 \text{ gives } A$$

$$12 = 3b^2 \;\Rightarrow\; b^2 = 4 \;\Rightarrow\; b = 2 \longleftarrow b > 0, \text{ so not } -2$$

$$y = 3 \times 2^x \longleftarrow \text{check: } 3 \times 2^2 = 12$$

Example: $y = A \times b^x$ passes through $(1, 6)$ and $(3, 54)$. Find A and b.

$$Ab = 6, \qquad Ab^3 = 54 \longleftarrow \text{one equation per point}$$

$$\frac{Ab^3}{Ab} = b^2 = 9 \;\Rightarrow\; b = 3 \longleftarrow \text{divide: } A \text{ cancels}$$

$$A = \frac{6}{3} = 2: \qquad y = 2 \times 3^x \longleftarrow \text{check: } 2 \times 3^3 = 54$$

RULE: half-life and doubling time

Half-life T: the time for the amount to halve. After time t: $y = A\left(\frac{1}{2}\right)^{t/T}$.

Doubling time T: the time for the amount to double. After time t: $y = A \times 2^{t/T}$.

t/T counts the halvings (or doublings). When the answer is a whole number of them, count; otherwise Chapter 22 solves it with logs.

Example: a substance has a half-life of 5 years. How much of 80 g is left after 20 years? When is 10 g left?

$$80 \times \left(\tfrac{1}{2}\right)^{20/5} = 80 \times \tfrac{1}{16} = 5 \text{ g}$$ ← *20 years = 4 half-lives*

$$80 \to 40 \to 20 \to 10 : \ 3 \text{ half-lives} = 15 \text{ years}$$

count the halvings

Example: a population of 500 doubles every 3 hours. When does it reach 8000?

$$\frac{8000}{500} = 16 = 2^4$$ ← *how many doublings?*

$$4 \times 3 = 12 \text{ hours}$$ ← *4 doublings of 3 hours*

Example: £2000 grows to £2420 after 2 years of compound interest. Find the yearly rate.

$$2000\,m^2 = 2420 \ \Rightarrow \ m^2 = 1.21 \ \Rightarrow \ m = 1.1$$

m = the yearly multiplier

$$m = 1 + \tfrac{r}{100} = 1.1 \ \Rightarrow \ r = 10\%$$

Example: a value is multiplied by 0.92 every year. What is the percentage change?

$0.92 = 1 - 0.08$, so it is a **decrease of 8%** a year.

Mistakes to avoid

✗ Common mistakes

✗ 5% rise: $\times 0.05$

the multiplier keeps the original 100%

✓ $\times 1.05$

✗ 20% fall: $\times 0.2$

1 − 0.20 = 0.80

✓ $\times 0.8$

✗ $1000 \times 1.05 \times 10$

multiply by 1.05 ten times: a power

✓ 1000×1.05^{10}

✗ $2 \times 3^x = 6^x$

only the 3 is raised to the power

✓ 2×3^x

✗ $b^2 = 9 \Rightarrow b = \pm 3$

the base of an exponential is positive

✓ $b = 3$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Exponential model	$y = A \times b^x$ A = start value (at $x = 0$); $b > 1$ growth, $0 < b < 1$ decay
e form	$y = Ae^{kx}$ $k > 0$ growth, $k < 0$ decay
Percentage change	$\times \left(1 \pm \dfrac{r}{100}\right)$ per step + for increase, − for decrease
Compound interest	$A\left(1 + \dfrac{r}{n}\right)^{nt}$ rate r as a decimal, n payments a year, t years
The number e	$e = \lim\limits_{n \to \infty} \left(1 + \dfrac{1}{n}\right)^n \approx 2.718$
Levelling off	$y = A + Be^{-kt} \ \rightarrow \ A$ starts at $A + B$; asymptote $y = A$
Time constant	$i = Ie^{-t/\tau}$ after τ, $e^{-1} \approx 37\%$ is left
Half-life	$y = A\left(\frac{1}{2}\right)^{t/T}$ T = half-life
Doubling time	$y = A \times 2^{t/T}$ T = doubling time
Two points	$x = 0$ gives A; otherwise divide the two equations to cancel A

Folding paper: your turn

Answers in grey. Use a calculator for the e questions.

Power or exponential?

1. $y = x^5$
2. $y = 5^x$
3. $y = 3 \times 0.8^x$
4. $y = 1/x^2$

1. power 2. exponential growth 3. exponential decay 4. power

Using $y = A \times b^x$

5. A colony of 200 bacteria doubles every hour. How many after 6 hours?
6. A £15,000 car loses 20% of its value each year. Value after 3 years?
7. £500 at 3% compound interest for 4 years?

5. $200 \times 2^6 = 12{,}800$ 6. $15{,}000 \times 0.8^3 = £7680$ 7. $500 \times 1.03^4 \approx £562.75$

e

8. Work out e^2 and e^{-2} (2 d.p.)
9. Using T = 20 + 70e^(−0.1t), find T when t = 5.
10. A current i = 4e^(−t/2) amps. Find i at t = 0 and t = 2.

8. 7.39 and 0.14 9. about 62.5 °C 10. 4 A and $4e^{-1} \approx 1.47$ A

Find the unknown

11. $y = A \times b^x$ passes through (0, 5) and (2, 45). Find A and b.
12. $y = A \times b^x$ passes through (1, 10) and (3, 40). Find A and b.
13. A drug has a half-life of 6 hours. How much of a 200 mg dose is left after 24 hours?
14. Bacteria double every 20 minutes. How long until 300 become 4800?
15. £5000 becomes £5618 after 2 years of compound interest. Find the yearly rate.
16. A car's value is multiplied by 0.85 each year. What is the yearly percentage change?

11. A = 5, b = 3 ($b^2 = 45/5 = 9$) 12. b = 2, A = 5 ($b^2 = 40/10 = 4$) 13. 12.5 mg (4 half-lives: $200 \times (1/2)^4$) 14. 80 minutes ($4800/300 = 16 = 2^4$) 15. 6% ($m^2 = 1.1236$, m = 1.06) 16. a 15% decrease (1 − 0.85)